

MINI PROJECT SYNOPSIS

ON

“SKILLCHAIN”

Smart Student & Placement Platform

SUBMITTED BY

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Declaration

We hereby declare that the project entitled “SkillChain – Smart Student & Placement Platform” is our own mini-project work, undertaken as part of our academic curriculum.

The project proposes a centralized platform where students can maintain profiles, skills, projects, certificates and resumes; administrators and faculty can verify credentials; and recruiters can post jobs, discover candidates and shortlist students. AI-assisted resume analysis and blockchain-backed credential verification are included as advanced features.

We further declare that the planning, design, development, testing and documentation of this project will be carried out by the team members listed below, under the guidance of Piyush Khandelwal.

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Abstract / Project Description

SkillChain is a full-stack web platform that brings student career records and campus placement activity into one system. It gives students a single place to build a profile, list skills and projects, upload certificates and a resume, and apply to jobs. Faculty and administrators use the same platform to verify credentials and manage placement activity, while recruiters use it to post openings, search verified candidates and shortlist applicants using AI-assisted match scores.

The application is built with a React.js frontend and a Node.js/Express.js backend, with MongoDB as the primary data store; JWT and bcrypt handle authentication and password security. Two advanced layers extend the core platform: an AI service that parses resumes, extracts and normalizes skills, and scores candidates against job requirements; and a lightweight blockchain layer that issues a tamper-evident proof for each verified certificate, checkable through a public, QR-enabled verification page. Only the certificate hash, issuer and timestamp are recorded on-chain — personal data and documents remain safely off-chain.

The project is developed in phases, starting with the core profile, job and application workflow, followed by role-based access control, and then the AI-matching and blockchain-verification features. The result is intended to be a working prototype that demonstrates a secure, centralized and verifiable alternative to today's scattered resumes, certificates and manual verification process.

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1. Introduction

SkillChain is a web-based, smart student and placement platform designed to unify student profiles, skills, projects, certificates and recruitment activity within a single responsive application. Students can build verified profiles and resumes, colleges and faculty can verify credentials, and recruiters can post jobs, search candidates and shortlist suitable students.

The platform is built on a React frontend with a Node.js and Express.js backend, backed by MongoDB for data storage. JWT and bcrypt secure authentication and passwords. Web3 components provide tamper-evident credential proof, while AI services support resume parsing, skill extraction and job matching.

About the Project

Through SkillChain, a student can create a profile, add skills and projects, upload a resume and apply for jobs. Recruiters can create job listings and discover candidates using skills and AI-generated match scores. Administrators or faculty can verify credentials, with each certificate represented by a cryptographic hash on a blockchain test network – private documents and personal information remain safely off-chain.

Project Title: “SkillChain – Smart Student & Placement Platform”

2. Primary Reason to Choose This Project

Students and recruiters currently depend on scattered resumes, certificates, job portals and manual verification processes, which makes candidate discovery, credential verification and skill comparison inefficient. SkillChain is proposed as a centralized digital solution for student profiles, placement workflows and credential verification. Key motivations include:

- A centralized online profile for skills, projects, certificates and resumes.
- Convenient job discovery and structured application management.
- Verified credentials that can be checked through a public verification page.
- Reduced dependence on scattered documents and manual verification.
- AI-assisted resume skill extraction and job matching.
- Recruiter search, filtering and candidate shortlisting.
- A practical full-stack use case for REST APIs, MongoDB and role-based access.
- A learning-oriented project covering Web3, AI, Docker, testing and CI/CD.

3. The Main Objective of the Project

The main objective of SkillChain is to develop a secure, centralized and user-friendly student and placement platform that will:

- Allow students to create and maintain profiles, skills, projects, certificates and resumes.
- Allow administrators/faculty to manage students and verify credentials.
- Allow recruiters to create jobs, search candidates and shortlist applicants.
- Provide AI-assisted resume parsing, skill extraction and job-match scoring.
- Provide blockchain-backed certificate verification and QR-enabled verification.
- Protect application data through authentication, authorization and input validation.
- Provide responsive dashboards on a maintainable, extensible architecture.

4. Scope of the Project

The scope of SkillChain covers digital student-profile management, placement workflows, credential verification and AI-assisted matching for a college-oriented ecosystem.

Student Scope

- Create and update profile information.
- Add skills and projects.
- Upload resumes and manage certificates.
- Browse jobs and submit applications.
- View application status and AI-assisted job-match information.

Administrator / Faculty Scope

- Manage student records and verification activities.
- Verify skills and certificates.
- Manage jobs and view activity/analytics.
- Maintain organized records for verification and placement.

Recruiter Scope

- Create and manage job postings.
- Search and filter student profiles.
- View verified profiles and shortlist candidates.
- Use AI match scores as an aid to candidate discovery.

The initial project focuses on the web platform and its core placement workflow. Advanced blockchain, AI, analytics, Docker, CI/CD and deployment capabilities will be added progressively once the MVP is stable.

5. Working Methodology of the Project

The development follows an iterative full-stack workflow. The team first identifies the student and

recruitment journeys and defines roles, profile information, skills, projects, certificates, jobs, applications and verification fields. Responsive React screens are then connected to Express REST APIs, MongoDB stores structured data, and each completed feature is tested with valid, invalid and boundary inputs before review.



Recommended learning / dependency chain: HTML/CSS/JavaScript → React → HTTP/REST → Node/Express → MongoDB → Authentication/RBAC → Full-stack workflows → Solidity basics → Hardhat → Ethers.js/MetaMask → AI integration → Docker/CI-CD → Deployment.

6. Details About the Hardware and Software

Category	Requirement
Operating System	Windows 10 / 11, Linux or macOS
Software Required	Visual Studio Code, Node.js, npm, MongoDB, Git & GitHub, a modern web browser, Postman
Frontend / Backend	React.js, Tailwind CSS, Node.js, Express.js
Blockchain / AI	Solidity, Hardhat, Ethers.js, MetaMask, LLM/API integration
Hardware	Dual-core or better processor; 4 GB RAM minimum, 8 GB recommended
Storage / Network	At least 10 GB free space and a stable internet connection

7. Testing Technology

- Postman – API and backend endpoint testing.
- Browser Developer Tools – frontend debugging and responsive-interface inspection.
- Manual functional testing – profile, job, application and credential workflows.
- Validation testing – missing fields, invalid inputs and file-upload checks.
- Role-based testing – verifying Student, Admin/Faculty and Recruiter permissions.
- Unit/API and frontend smoke tests during the quality phase.

8. Frontend and Backend

Frontend

The frontend of SkillChain will be developed using HTML, CSS, JavaScript, React.js, React Router and Tailwind CSS. It will include a landing page, authentication screens, student dashboard, recruiter dashboard, administrator/faculty dashboard, profile and skill pages, project and certificate pages, job listings, applications, AI match views and a public credential-verification page. The frontend is responsible for clear navigation, responsive presentation and client-side validation.

Backend

The backend will be developed using Node.js, Express.js, MongoDB, Mongoose, JWT authentication, bcrypt and REST APIs. It will manage users, skills, projects, certificates, jobs, applications, resumes, verification status and role-based access. All requests will be routed through the backend so that API secrets remain protected.

Application Architecture

React communicates with the Express API through well-defined REST endpoints. The API validates incoming data, applies access rules and persists approved records in MongoDB. MetaMask and Ethers.js support wallet operations, while blockchain transactions provide credential proof. This separation keeps user-interface concerns independent of business logic and makes the platform easier to test and extend.

Quality and Security Considerations

Administrative routes are protected through authenticated access, passwords are securely hashed, server-side validation is enforced, private keys are never exposed to the frontend, and sensitive documents are kept off the public blockchain.

9. Module Description

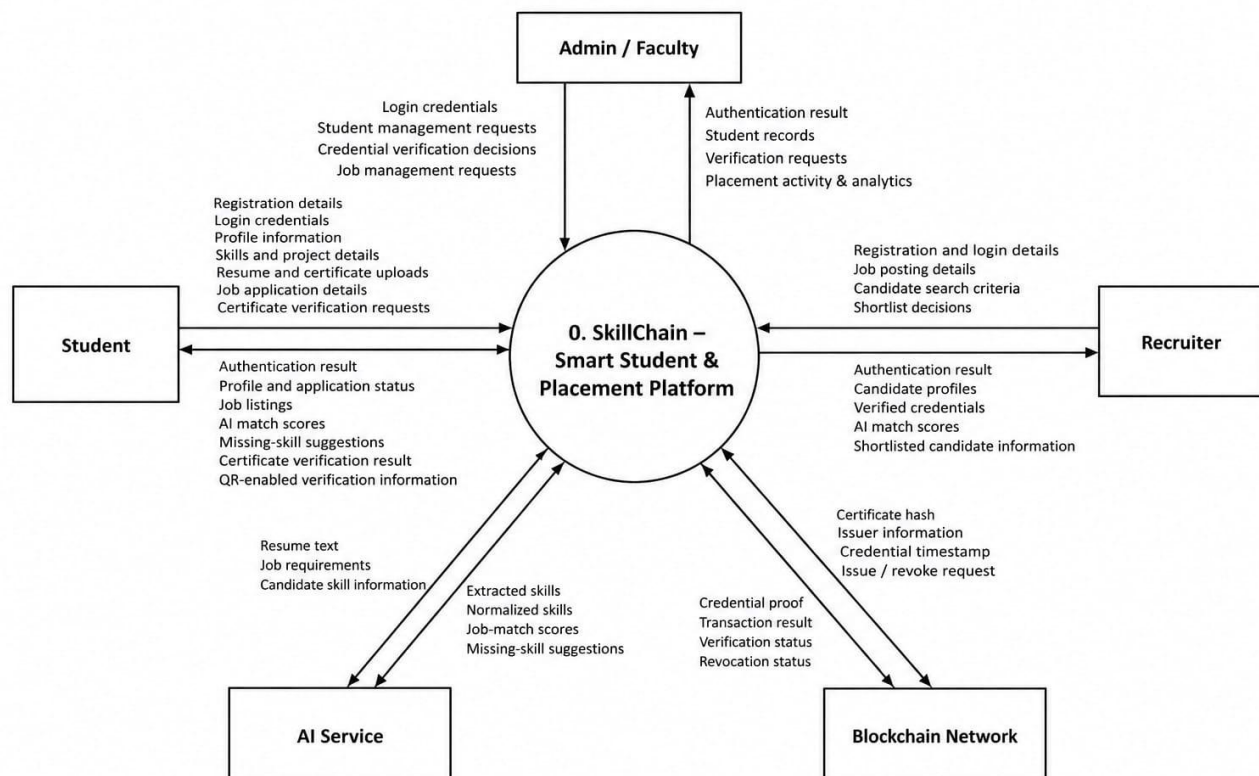
Module	Description
Student Profile Module	Maintains student profile, skills, projects, certificates and resume information.
Job & Application Module	Allows recruiters to create jobs and students to discover and apply for jobs.
Recruiter Module	Provides candidate search, filters, verified profiles and shortlisting.
Administrator Module	Manages students, credential verification, jobs and analytics.
Authentication & Access Module	Protects accounts and privileged routes using JWT, bcrypt and RBAC.
Resume & AI Module	Uploads resumes, extracts skills, and produces job-match scores with missing-skill suggestions.
Credential Verification Module	Creates certificate proof, stores credential hashes on-chain, and supports verification/revocation.
QR & Public Verification Module	Provides QR-enabled access to a public certificate-verification page.

10. Data Flow Diagrams

0-Level DFD – Context Diagram

At the context level, Student, Recruiter and Admin/Faculty actors interact with the central SkillChain platform, which also interfaces directly with the AI Service and the Blockchain Network for skill matching and credential verification.

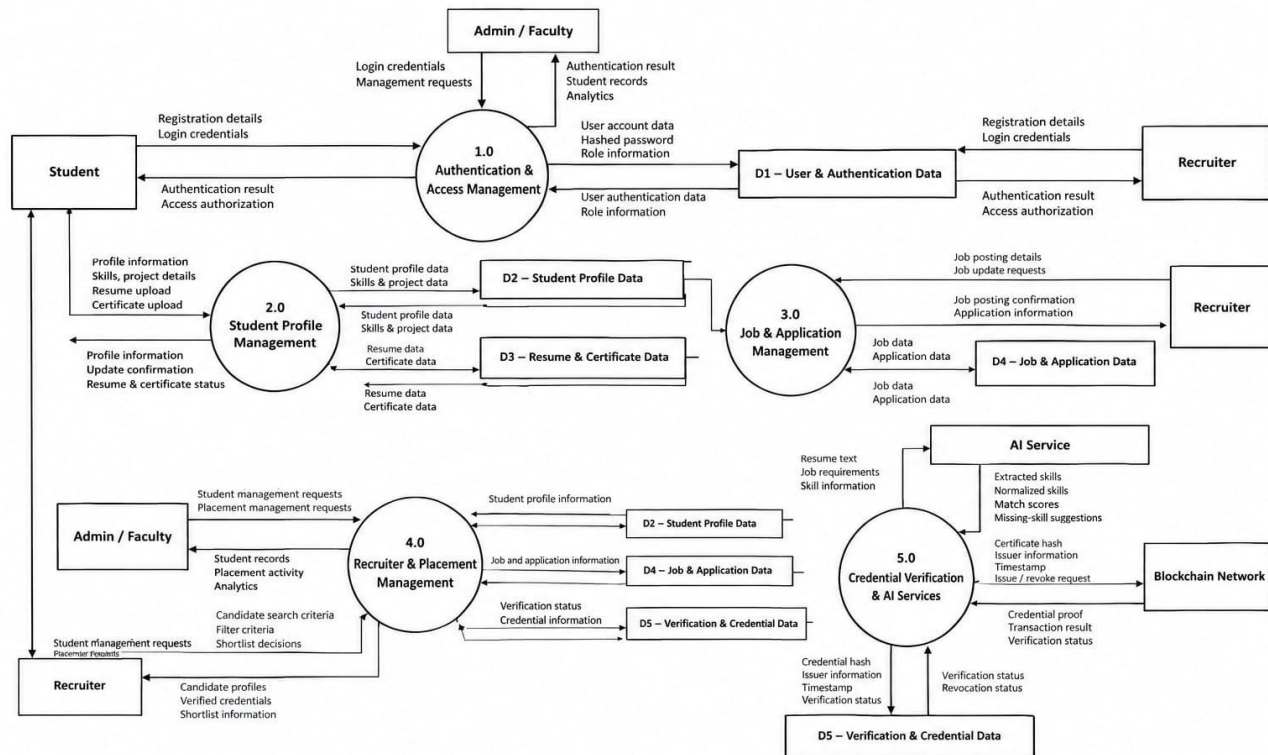
0-Level DFD – Context Diagram



1-Level DFD – System Decomposition

The Level-1 diagram decomposes the platform into five core processes – Authentication & Access Management, Student Profile Management, Job & Application Management, Recruiter & Placement Management, and Credential Verification & AI Services – each reading from and writing to its own data store (D1–D5).

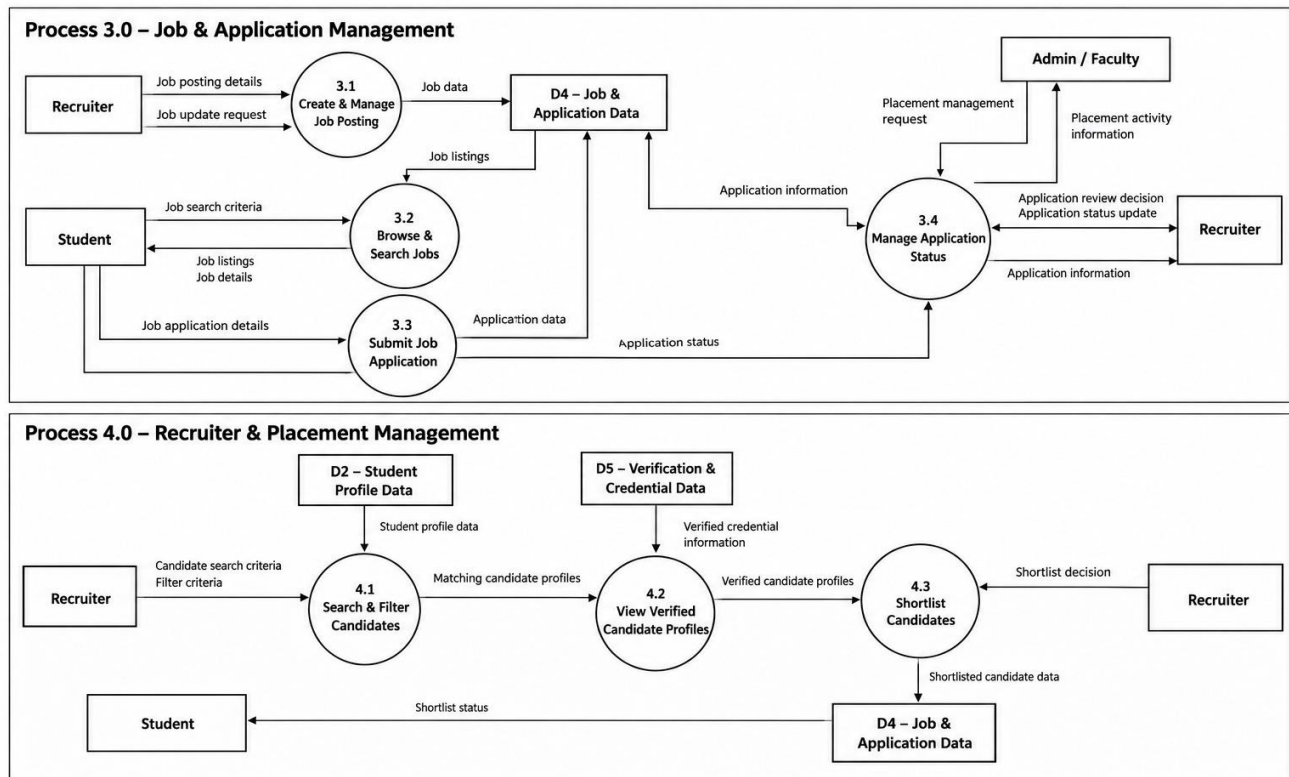
1-Level DFD – SkillChain System Decomposition



2-Level DFD – Job Application & Recruitment Workflow

This diagram expands Process 3.0 (Job & Application Management) and Process 4.0 (Recruiter & Placement Management) into their component sub-processes: creating and managing job postings, browsing and searching jobs, submitting applications, managing application status, and searching, filtering and shortlisting verified candidates.

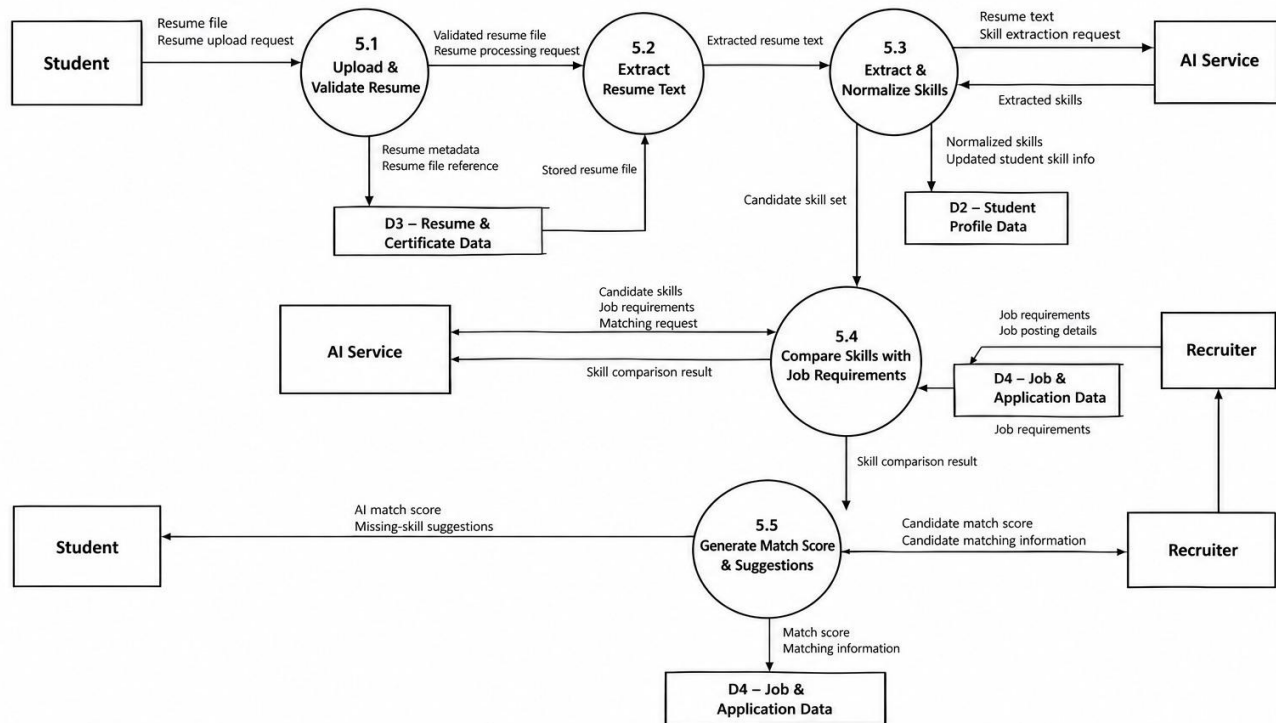
2-Level DFD – Job Application & Recruitment Workflow



2-Level DFD – Resume Processing & AI Job Matching

This diagram decomposes Process 5.0 into the resume-and-matching pipeline: uploading and validating the resume, extracting resume text, extracting and normalizing skills via the AI Service, comparing candidate skills with job requirements, and generating a match score with missing-skill suggestions.

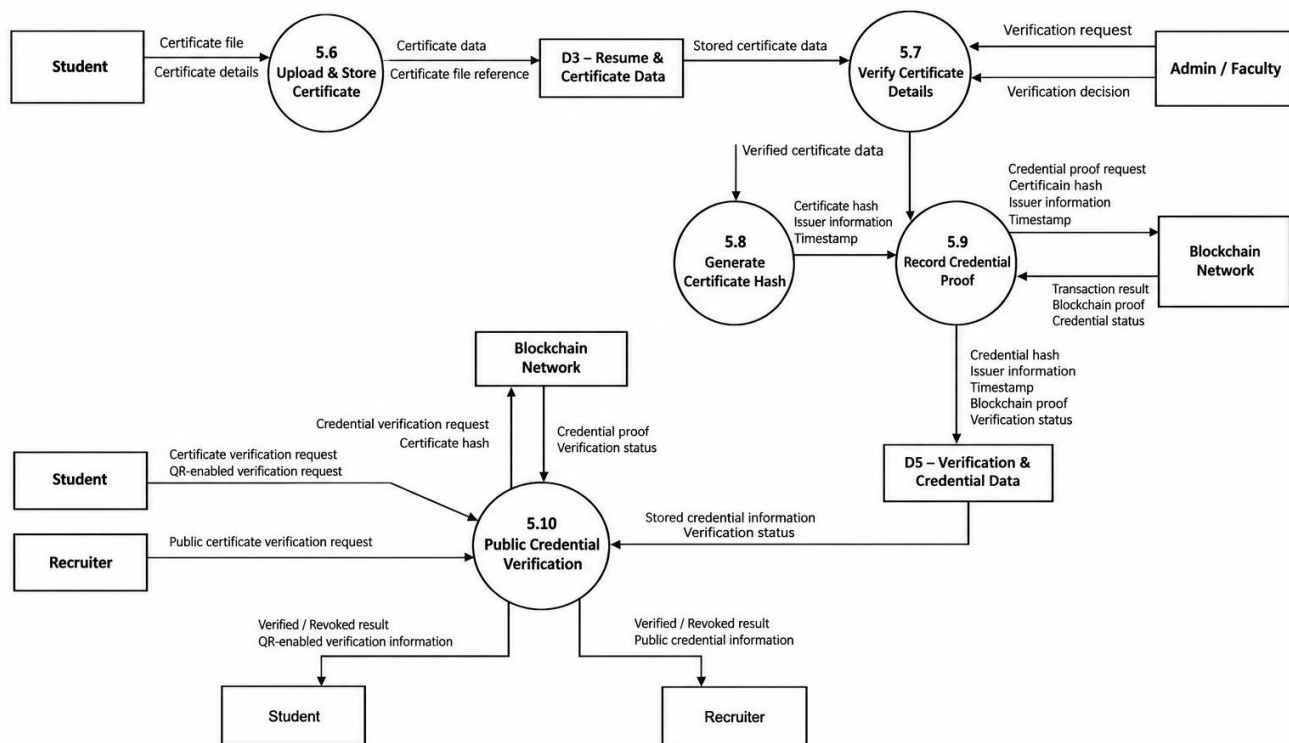
2-Level DFD – Resume Processing & AI Job Matching (Decomposition of Process 5.0 – Credential Verification & AI Services)



2-Level DFD – Credential Verification & Blockchain Proof

This diagram decomposes the remainder of Process 5.0: uploading and storing a certificate, verifying certificate details, generating a certificate hash, recording credential proof on the Blockchain Network, and supporting public, QR-enabled credential verification for students and recruiters.

2-Level DFD – Credential Verification & Blockchain Proof



Final Demo Flow

1. Student registers and logs in.
2. Student completes profile and uploads resume.
3. AI extracts skills and displays a job-match score.
4. Student adds a project and applies for a job.
5. Admin/Faculty verifies a certificate.
6. Backend hashes certificate data and records proof on a blockchain test network.
7. Student receives a QR-enabled certificate.
8. Recruiter opens the public verification page.
9. System compares the certificate proof and displays Verified or Revoked.

11. Web3 and AI Features

Web3 / Blockchain

The smart contract will be intentionally small, with three core operations: `issueCredential`, `revokeCredential` and `verifyCredential`. The system stores only the proof/metadata needed for verification – credential hash, issuer and timestamp – while private student data and PDF documents remain off-chain.

Artificial Intelligence

The AI feature extracts skills from resume text, normalizes them against the platform skill database, compares candidate skills with job requirements, generates a match score and provides missing-skill suggestions.

12. Security Considerations

- Hash passwords and never store plaintext passwords.
- Use JWT securely and protect privileged routes.
- Validate and sanitize all API inputs.
- Use environment variables for database, AI and blockchain secrets.
- Never place private keys in frontend code or GitHub.
- Apply least-privilege roles within the smart contract.
- Do not store sensitive personal documents on a public blockchain.
- Add rate limiting to login and AI endpoints.
- Check uploaded file type and size.
- Use HTTPS in production.

13. Expected Outcomes, Future Scope and Conclusion

Expected Outcomes

The expected outcome is a working SkillChain prototype featuring role-based dashboards, persistent student and recruitment data, secure authentication, resume handling, AI-assisted job matching and blockchain-backed certificate verification. The final project will also include documentation, testing results and a deployment-ready repository.

Future Scope

Future enhancements may include more granular permissions, optimized queries, refresh/session hardening, richer on-chain events, advanced AI recommendations, public audit history, Docker-based deployment, CI/CD pipelines and system monitoring.

Conclusion

SkillChain combines student profile management, placement workflows, AI assistance and verifiable credentials within a single platform. By cleanly separating application data, backend business logic, AI services and blockchain proof, the system delivers a practical B.Tech mini-project while covering modern full-stack development, Web3, AI, security, testing and deployment concepts.

14. References

1. React Documentation – React components, hooks and frontend development.
2. Node.js Documentation – backend runtime and server-side JavaScript.
3. Express.js Documentation – REST APIs, routing and middleware.
4. MongoDB Documentation – database design and data management.
5. Mongoose Documentation – MongoDB object modeling and schemas.
6. Solidity / Hardhat / Ethers.js Documentation – smart-contract development and blockchain integration.